Equivalent fractions

1 Select the equivalent fractions from the following lists:

a $\frac{4}{40}, \frac{1}{10}, \frac{2}{80}, \frac{8}{80}, \frac{8}{20}$

b $\frac{20}{30}, \frac{7}{3}, \frac{8}{12}, \frac{2}{6}, \frac{2}{3}$

c $\frac{1}{4}, \frac{16}{64}, \frac{4}{8}, \frac{128}{256}, \frac{32}{64}$

d $\frac{9}{8}, \frac{27}{24}, \frac{108}{96}, \frac{8}{9}, \frac{54}{24}$

e $\frac{15}{20}, \frac{6}{8}, \frac{10}{15}, \frac{9}{12}, \frac{2}{3}$

f $\frac{6}{9}, \frac{3}{4}, \frac{10}{15}, \frac{17}{24}, \frac{4}{6}$

g $\frac{4}{18}, \frac{14}{63}, \frac{2}{16}, \frac{2}{9}, \frac{10}{36}$

h $\frac{17}{5}, \frac{8}{20}, \frac{2}{5}, \frac{4}{14}, \frac{16}{40}$

2 Complete the following equivalent fraction statements:

a $\frac{1}{2} = \frac{\square}{4}$

b $\frac{1}{5} = \frac{\square}{75}$

c $\frac{3}{8} = \frac{\square}{40}$

d $\frac{9}{5} = \frac{\square}{40}$

e $\frac{19}{10} = \frac{\square}{90}$

f $\frac{3}{8} = \frac{\square}{24}$

g $\frac{5}{16} = \frac{20}{\square}$

h $\frac{7}{12} = \frac{35}{\square}$

i $\frac{4}{9} = \frac{40}{\square}$

j $\frac{4}{3} = \frac{48}{\square}$

k $\frac{5}{21} = \frac{45}{\square}$

l $\frac{18}{11} = \frac{72}{\square}$

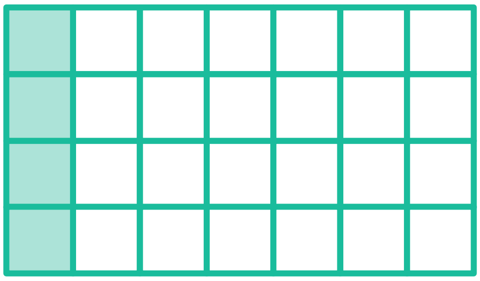
3 Complete the table below:

Mixed number	Improper fraction	Equivalent fraction
$2\frac{1}{6}$		
$3\frac{4}{5}$		
$4\frac{2}{7}$		
$1\frac{2}{5}$		
$6\frac{1}{8}$		
$2\frac{3}{8}$		
$3\frac{3}{7}$		
$4\frac{2}{9}$		
$5\frac{1}{4}$		

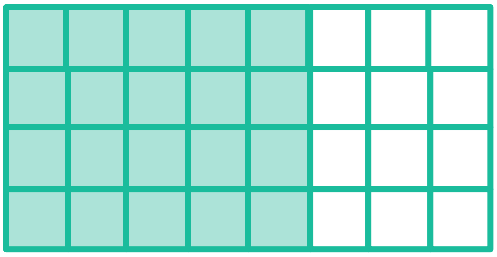
Simplify fractions

4 Use the diagrams provided to simplify the following fractions:

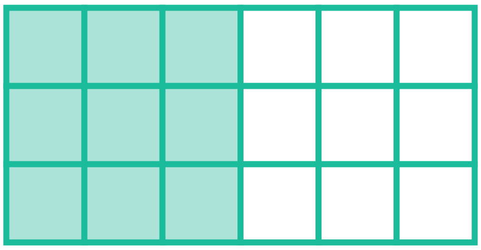
a $\frac{4}{28}$



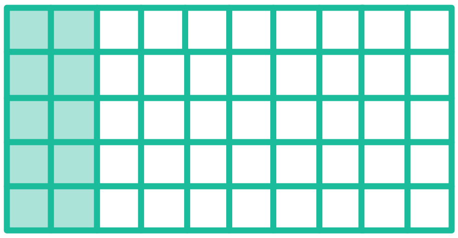
b $\frac{20}{32}$



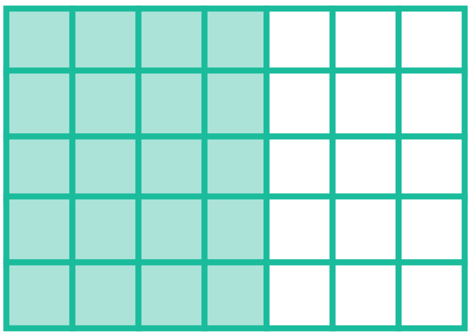
c $\frac{9}{18}$



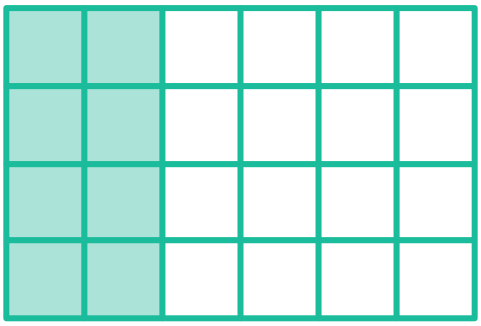
d $\frac{10}{50}$



e $\frac{20}{35}$

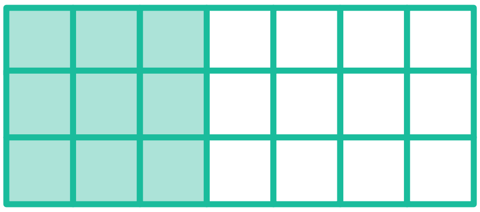


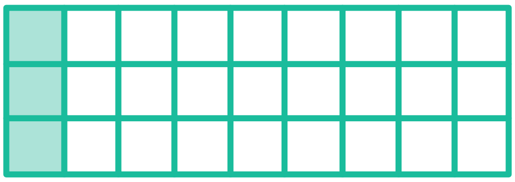
f $\frac{8}{24}$



g $\frac{9}{21}$

h $\frac{3}{27}$





5 Simplify the following fractions:

a $\frac{3}{9}$ b $\frac{20}{60}$ c $\frac{6}{20}$ d $\frac{27}{45}$

e $\frac{63}{28}$ f $\frac{40}{30}$ g $\frac{26}{50}$ h $\frac{24}{66}$

i $\frac{32}{80}$ j $\frac{64}{92}$ k $\frac{80}{100}$ l $\frac{60}{200}$

m $\frac{2}{10}$ n $\frac{8}{64}$ o $\frac{21}{231}$ p $\frac{150}{270}$

Compare fractions

6 For the each of the following pairs of fractions, state which fraction is bigger:

a $\frac{1}{3}, \frac{3}{15}$ b $\frac{4}{28}, \frac{6}{14}$ c $\frac{31}{8}, \frac{24}{7}$ d $\frac{3}{8}, \frac{1}{2}$

e $\frac{14}{5}, \frac{11}{4}$ f $\frac{5}{2}, \frac{49}{20}$ g $1\frac{6}{7}, \frac{25}{14}$ h $3\frac{4}{9}, \frac{7}{2}$

7 For the each of the following pairs of fractions, state which fraction is smaller:

a $\frac{2}{9}, \frac{7}{8}$ b $\frac{7}{5}, \frac{8}{10}$ c $\frac{4}{15}, \frac{2}{5}$ d $\frac{4}{7}, \frac{4}{8}$

e $\frac{63}{15}, \frac{82}{20}$ f $\frac{21}{6}, \frac{18}{5}$ g $\frac{27}{45}, \frac{5}{8}$ h $2\frac{5}{9}, 2\frac{7}{11}$

8 For each of the following groups of fractions:

i Find the lowest common denominator of the 3 fractions.

ii Rewrite all 3 fractions with this lowest common denominator.

iii List the 3 fractions in ascending order.

a $\frac{2}{5}, \frac{9}{55}, \frac{7}{11}$ b $\frac{7}{13}, \frac{8}{65}, \frac{3}{5}$ c $\frac{5}{7}, \frac{9}{77}, \frac{8}{11}$ d $\frac{5}{17}, \frac{7}{68}, \frac{3}{4}$

9 Arrange the following groups of fractions from smallest to largest:

a $\frac{11}{20}, \frac{4}{15}, \frac{2}{5}$ b $\frac{2}{3}, \frac{3}{7}, \frac{4}{5}$ c $\frac{21}{9}, \frac{6}{4}, \frac{2}{7}$ d $4\frac{1}{2}, \frac{31}{4}, \frac{15}{2}$

e $\frac{39}{4}, 4\frac{5}{10}, \frac{19}{2}$ f $1\frac{6}{7}, \frac{33}{14}, \frac{16}{7}$ g $2\frac{2}{5}, \frac{33}{10}, \frac{16}{5}$ h $\frac{17}{3}, \frac{35}{6}, 3\frac{2}{3}$

10 For each of the following groups of fractions:

i Find the lowest common denominator of the 3 fractions.

ii Rewrite all 3 fractions with this lowest common denominator.

iii List the 3 fractions in descending order.

a $\frac{1}{4}, \frac{3}{8}, \frac{5}{32}$ b $\frac{2}{6}, \frac{4}{9}, \frac{1}{54}$ c $\frac{2}{5}, \frac{6}{7}, \frac{8}{35}$ d $2\frac{2}{5}, \frac{37}{10}, \frac{18}{5}$

11 Arrange the following groups of fractions from largest to smallest:

a $\frac{1}{2}, \frac{2}{3}, \frac{5}{8}$ b $\frac{3}{8}, \frac{9}{16}, \frac{13}{24}$ c $2\frac{7}{10}, \frac{13}{5}, \frac{7}{2}$ d $2\frac{1}{7}, \frac{33}{14}, \frac{16}{7}$

e $6\frac{1}{2}, \frac{35}{4}, \frac{17}{2}$ f $\frac{33}{10}, 1\frac{4}{5}, \frac{16}{5}$ g $\frac{17}{4}, 4\frac{3}{4}, \frac{39}{8}$ h $\frac{21}{2}, \frac{37}{4}, 9\frac{1}{2}$